Terminal (Airport / Industrial / Air Operations & Ground Warfare)

Executive Summary

Terminal is a massive **airport+industrial battlefield** built for **air operations, runway control, ground warfare, logistics disruption, and multi-domain coordination**. The terrain features runways, taxiways, hangars, cargo terminals, fuel farms, radar towers, maintenance bays, and elevated service roads. The map emphasizes Obsidian Protocol's autonomous air-ground coordination, sensor-network routing, comms-relay warfare, and logistics under high-traffic industrial constraints. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10×10 m physical area representing ~1 km² of airport territory. Key features include runway chokepoints, hangar interior combat, airspace corridors, cargo-yard flanking routes, and strategic control of radar and comms infrastructure.

Design Goals & Gameplay

Air Operations & Runway Control
Terminal prioritizes aerial combat, runway denial, VTOL deployment, drone reconnaissance, and air-ground coordination. Controlling runways determines air superiority.
Industrial Ground Warfare
Cargo yards, hangars, and maintenance bays create mixed-range combat zones. Long fire lanes contrast with tight interior routes.
Autonomy & Multi-Domain Coordination
Autonomous units coordinate air and ground forces, selecting optimal routes based on runway status, radar coverage, and enemy movement.
Sensor & Information Warfare
Heat shimmer, exhaust, smoke, and radar interference distort sensors. Relay towers on control towers restore clarity. Destroying relays creates blind pockets across the airport grid.
Logistics & Resources
Fuel farms, cargo terminals, and maintenance depots serve as resupply nodes. Convoys must navigate exposed runways and avoid ambush-prone industrial corridors.
Deployment & Progress
A fixed deployment budget (~10,000 points) forces players to choose between air support, heavy armor, or autonomous infantry squads. Blue deploys from the southwest cargo apron; Red deploys from the northeast control tower stronghold.

Spatial Constraints (Nreal AR)

Device Capabilities
Nreal Light/Air: ~52° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.
Playable Area & Scaling
10×10 m physical area = 1 km² airport zone (1:100 scale). Designed for tabletop AR.
Tracking & Occlusion
Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System
1000×1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.
Key Locations
Blue HQ (Cargo Apron Command) — (0,0). 150×150 m Cargo loaders, forklifts, drone pads, armored station.
Red HQ (Control Tower Stronghold) — (1000,1000). 150×150 m Radar mast. AA emplacements, hardened command center.
Central Runway Complex — (500,450). 350×200 m Two runways, taxiways, airspace corridors, long-range fire lanes.
Cargo Terminal & Rail Spur — (700,800). 150×150 m Cargo containers, cranes, rail spurs, flanking routes.
Hangar Row — (800,300). 200×100 m Interior combat, maintenance bays, aircraft shelters.
Fuel Farm — (l.450 → 400,900) Fuel tanks, pipelines, hazardous choke point.
Service Road Route — (100,0 → 450,150) Safer ground route with partial cover.
Radar Peak A — (450,150). +260 m Long-range sensor vantage point.
Relay Tower B — (150,650). +280 m Comms relay tower and drone hub.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Runway Complex
Extreme long-range sightlines; controlling runways determines air dominance.
Hangar Row
Interior flanking routes; close-range engagements.
Fuel Farm
Hazardous chokepoint; explosions and fire risk.
Cargo Terminal
Industrial cover; ideal for ambushes and flanking.
Sightlines
Radar Peak A provides long-range sensor coverage (~350 m). Heat shimmer and smoke create fog-of-war pockets.

Deployment Zones

Blue Deployment: Cargo apron + ground station
Red Deployment: NE control tower stronghold + AA emplacements.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain
Runways, taxiways, hangars, cargo yards, fuel farms, service roads.
Vegetation
Sparse shrubs, grass strips, airport landscaping.
Buildings & Props
Military: AA guns, radar dishes, watchtowers. Industrial: Hangars, cranes, pipelines, generators. Logistics: Trucks, cargo containers, fuel drums, drones.
Effects
Smoke, sparks, dust, heat shimmer, exhaust plumes.
Audio
Aircraft noise, machinery, radio chatter, wind.

LOD & Budget

Asset Cate	Count	Tri/Item	Total Tris
Runways/	3	200k	600k
Hangars/I	4	80k	320k
Buildings/	15	50k	750k
Roads/Pa	4	30k	120k
Vegetatio	120	2k	240k
Props	50	10k	500k
Total	—	—	**~2.52M**

Texture Budget
2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins
Unreal ARTemplate + Nreal SDK (NREAL). ARKit/ARCore enabled.
Rendering
Forward renderer, stationary lights, baked GI, aggressive culling.
Anchors
Single spatial anchor for entire map.
Interaction
Phone pointer for pan/zoom; HUD as screen-space widgets.
Multiplayer
Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones
1. Prototype Layout — Month 1 Block-out runways, hangars, cargo terminal. AR anchor test.
2. Core Systems — Month 2 NavMesh, air-ground autonomy behaviors.
3. AR Integration — Month 3 Anchors, plane detection, basic interactivity.
4. Content Pass — Month 4–5 Final models, textures, lighting, audio.
5. Gameplay & Polish — Month 6–7 Objectives, balance, VFX (heat shimmer, smoke, sparks).
6. Testing & QA — Month 8 Multiplayer stress test, sensor validation.

Testing Plan

Autonomy
Verify air-ground coordination, runway logic, hangar navigation.
Comms/Sensors
Test relay destruction, fog-of-war updates, LOS checks.
Performance
≥60 Hz AR render on target device.
QA Checklist
Navigation, combined-arms behavior, deployment budget enforcement, anchor drift, sensor accuracy.